

Bulan 5: Security & Performance

Minggu 1: Security Best Practices

Hari 1: CSRF & XSS Protection

Project: SecureForm

- Deskripsi: Form security demonstration dengan attack simulation
- Fitur: CSRF tokens, XSS sanitization, CSP headers
- Keunikan: Interactive security test lab

Hari 2: SQL Injection Prevention

Project: SafeQuery

- Deskripsi: Query builder dengan injection prevention
- Fitur: Parameterized queries, input validation, security audit
- Keunikan: Vulnerability scanner dengan fix suggestions

Hari 3: Authentication Security

Project: AuthShield

- Deskripsi: Hardened authentication dengan brute-force protection
- Fitur: Rate limiting, account lockout, 2FA, security questions
- Keunikan: Behavioral biometrics (typing patterns)

Hari 4: Permission & Authorization

Project: PermissionVault

- Deskripsi: Advanced RBAC dengan attribute-based control
- Fitur: Role hierarchy, resource permissions, audit trail
- Keunikan: Dynamic permission evaluation engine

Hari 5: Data Encryption

Project: CryptoBox

- Deskripsi: Field-level encryption untuk sensitive data

- Fitur: Encrypted fields, key rotation, HSM integration
- Keunikan: Zero-knowledge architecture simulation

Hari 6: API Security

Project: APIGuard

- Deskripsi: API security suite dengan threat detection
- Fitur: API keys, OAuth2, rate limiting, IP whitelist
- Keunikan: ML-based anomaly detection

Hari 7: Security Headers & HTTPS

Project: HeaderShield

- Deskripsi: Security headers configurator dengan SSL
- Fitur: HSTS, CSP, CORS, referrer policy
- Keunikan: Security score calculator dengan recommendations

Minggu 2: Performance Optimization

Hari 8: Database Query Optimization

Project: QueryBoost

- Deskripsi: Automated query optimizer dengan benchmarking
- Fitur: Index suggestions, query rewriting, explain analyze
- Keunikan: AI-powered query optimization

Hari 9: Caching Strategies Advanced

Project: CacheMaster

- Deskripsi: Multi-tier caching dengan intelligent invalidation
- Fitur: Redis, Memcached, CDN integration, cache warming
- Keunikan: Predictive cache prefetching

Hari 10: Template Optimization

Project: TemplateSpeed

- Deskripsi: Template rendering optimization toolkit

- Fitur: Fragment caching, lazy loading, minification
- Keunikan: Template performance profiler

Hari 11: Static Assets Optimization

Project: AssetPipeline

- Deskripsi: Asset optimization dengan webpack/vite
- Fitur: Minification, compression, bundling, tree-shaking
- Keunikan: Automatic code splitting based on routes

Hari 12: Database Connection Pooling

Project: PoolManager

- Deskripsi: Connection pool optimizer dengan monitoring
- Fitur: pgBouncer integration, pool sizing, connection lifecycle
- Keunikan: Auto-scaling pool based on load

Hari 13: Lazy Loading & Pagination

Project: InfiniteView

- Deskripsi: Efficient data loading dengan virtual scrolling
- Fitur: Cursor pagination, prefetching, skeleton loaders
- Keunikan: Adaptive pagination based on network speed

Hari 14: Load Testing & Benchmarking

Project: LoadTest

- Deskripsi: Load testing suite dengan Locust integration
- Fitur: Stress testing, spike testing, endurance testing
- Keunikan: Realistic user behavior simulation

Minggu 3: Scalability & Architecture

Hari 15: Microservices Basics

Project: ServiceMesh

- Deskripsi: Microservices architecture dengan service discovery
- Fitur: Service registry, API gateway, inter-service communication
- Keunikan: Circuit breaker pattern implementation

Hari 16: Message Queues

Project: MessageBroker

- Deskripsi: Event-driven architecture dengan RabbitMQ
- Fitur: Pub/sub, request/reply, dead letter queue
- Keunikan: Message replay untuk debugging

Hari 17: Redis Advanced Usage

Project: RedisToolkit

- Deskripsi: Redis untuk caching, sessions, pub/sub
- Fitur: Sorted sets, HyperLogLog, geospatial
- Keunikan: Redis Streams untuk event sourcing

Hari 18: Database Sharding

Project: ShardRouter

- Deskripsi: Horizontal partitioning implementation
- Fitur: Hash-based sharding, range sharding, rebalancing
- Keunikan: Automatic shard migration

Hari 19: Load Balancing

Project: LoadBalance

- Deskripsi: Load balancer configuration dengan nginx
- Fitur: Round-robin, least connections, health checks
- Keunikan: Dynamic upstream configuration

Hari 20: Containerization (Docker)

Project: DockerStack

- Deskripsi: Docker compose setup untuk Django stack

- Fitur: Multi-container orchestration, volumes, networking
- Keunikan: Development/production parity

Hari 21: Kubernetes Basics

Project: K8sCluster

- Deskripsi: Django deployment pada Kubernetes
- Fitur: Pods, services, deployments, auto-scaling
- Keunikan: Blue-green deployment automation

Minggu 4: Monitoring & DevOps

Hari 22: Application Monitoring

Project: MonitorHub

- Deskripsi: APM dengan Sentry/New Relic integration
- Fitur: Error tracking, performance monitoring, alerting
- Keunikan: Predictive failure detection

Hari 23: Logging & Log Aggregation

Project: LogStream

- Deskripsi: Centralized logging dengan ELK stack
- Fitur: Log parsing, indexing, visualization, alerting
- Keunikan: Log anomaly detection dengan ML

Hari 24: Health Checks & Metrics

Project: HealthCheck

- Deskripsi: Comprehensive health monitoring system
- Fitur: Database, cache, queue health, custom checks
- Keunikan: Self-healing mechanisms

Hari 25: CI/CD Pipeline

Project: DeployFlow

- Deskripsi: Automated deployment pipeline dengan GitHub Actions

- Fitur: Build, test, deploy, rollback automation
- Keunikan: Canary deployments dengan automatic rollback

Hari 26: Database Backup & Recovery

Project: BackupPro

- Deskripsi: Automated backup solution dengan point-in-time recovery
- Fitur: Scheduled backups, encryption, offsite storage
- Keunikan: Automated disaster recovery testing

Hari 27: Infrastructure as Code

Project: TerraformDjango

- Deskripsi: Infrastructure provisioning dengan Terraform
- Fitur: AWS resources, auto-scaling, RDS, S3
- Keunikan: Cost optimization analyzer

Hari 28: Zero-Downtime Deployment

Project: BlueGreen

- Deskripsi: Zero-downtime deployment strategies
- Fitur: Blue-green, canary, rolling updates
- Keunikan: Automated rollback based on metrics

Hari 29-30: Production-Ready Project

Project: SaaS Starter

- Deskripsi: Production-ready SaaS boilerplate
- Fitur: Multi-tenancy, subscriptions, admin panel, monitoring, CI/CD, security hardening, performance optimizations, documentation
- Keunikan: One-command deployment, tenant isolation, auto-scaling
- Scope: 2 hari untuk enterprise-grade boilerplate